

Vascular surgery for penetrating injury of the neck*

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Summary. *Background:* Vascular injuries due to penetrating neck trauma are rare, but potentially lethal injuries.

Methods: 17 patients were treated for vascular injuries after penetrating neck trauma by specialised vascular surgeons between 1.1.1998 and 31.12.2010. All patients underwent surgical exploration and vascular surgical treatment for bleeding.

Results: Trauma mechanism was stab wound in 9 patients, metal fragments in 3 and various items in 5 patients. 4 patients had attempted suicide, and 3 patients were injured in a fight or murder attempt. The injuries were to zone II according to Monson in 14 patients, zone I in 2 patients, and zones I and II in 1 patient. In 8 patients arterial injuries and in 9 patients venous lesions were treated. Mortality was 11.8% (2/17 patients).

Conclusions: Penetrating trauma to the neck is a serious and potentially lethal injury. CTA is recommended in the literature in diagnostic exploration of hemodynamically stable patients. Emergency surgery by an experienced vascular surgeon can help to determine the extent of trauma and its treatment. In arterial trauma to the carotid arteries a complete intraoperative exposure with external and internal inspection of the vessel including the intimal layer is recommended. The role of endovascular treatment for vascular injury due to penetrating neck trauma needs further evaluation.

Keywords: Penetrating neck trauma, vascular injury, vascular reconstruction.

present a diagnostic and therapeutic challenge. For civilians with penetrating neck injuries the current mortality ranges from 3 to 6% [1]. Aim of this study was to analyze the results of our current diagnostic and operative strategies for penetrating neck trauma in a civilian setting.

Patients and methods

In a specialised vascular surgery database patient and vascular surgery data were collected prospectively. Retrospectively the records of all patients undergoing vascular surgery for non-iatrogenic traumatic lesions between 1.1.1998 and 31.12.2010 were analyzed. An experienced vascular surgeon is always in-hospital and present at first admission of any major trauma patient in our hospital.

Data regarding patient demographics, mechanism of injury and clinical presentation were retrieved from the database and entered in a new Excel file. Operation notes as well as photographs documented the vascular lesion, method of repair and associated injuries. Initial management and resuscitation were according to standard advanced trauma and life support (ATLS®) guidelines. Operative repair was according to vascular surgical emergency standards [2].

Results of the retrospective analysis are presented as mean±standard error of the mean (SEM). Descriptive analysis was performed for clinical and demographic data. Patients treated for iatrogenic vascular injuries were excluded from the analysis. Additionally patients operated by other departments for penetrating neck lesions without vascular trauma are not included in this report.

Introduction

Although penetrating neck injuries are rare in vascular surgical practice with 0.3 to 4% of procedures, they

Results

A total of 80 patients underwent vascular surgical operations for penetrating trauma between 1.1.1998 and 31.12.2010. Surgery for vascular lesions due to penetrating trauma of the neck was performed in 17 patients (17/80 = 31.3%) between 20 and 79 years with a mean age of 45.8 ± 19.6 years. Two patients of the 14 male and 3 female patients were polytraumatized.

Trauma mechanism was penetrating injury by a knife in 9 patients, by metal car fragments in 3 patients,

*Dedicated to Professor Karlheinz Tscheliessnigg on the occasion of his 65th birthday

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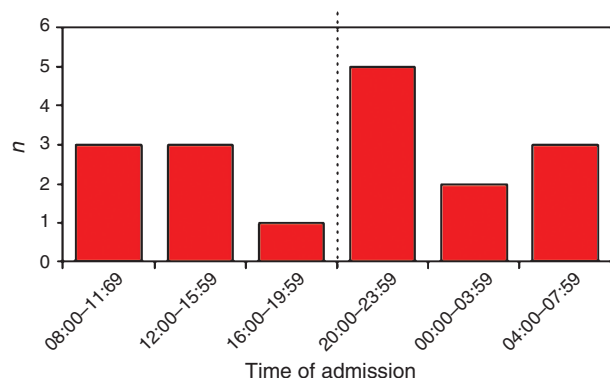


Fig. 1: Time of first patient admission to the hospital after penetrating neck injury (n = 17 patients)

and in one patient each by glass, a twig, a chain saw, a razor blade and a staple gun. Injuries following attempted suicide were treated in 4 patients. In 3 patients a second person had injured the patient during a fight or a murder attempt.

Of our 17 patients, 10 (58.8%) were admitted after regular hospital working hours (Fig. 1). By helicopter, 9 patients were transported to the hospital; by emergency van including trained emergency physician (“Notarztwagen”), 8 patients were admitted.

The majority of patients were taken from the emergency department directly to the operating room (12/17 pts. ≥70.6%). In 5 patients a CT-angiography was performed preoperatively, and in one patient an additional arterial angiography.

Indication for surgery was bleeding in all patients. The vascular surgeon treated lesions of the arterial system in 8 patients ($8/17 = 47.6\%$) and of the venous system in 9 patients ($9/17 = 52.4\%$). Surprisingly, no patient suffered from additional injuries involving the trachea, the esophagus or major nerves. The injuries were located in zone II in 14 patients, zone I in 2 patients and in zones I and II in one patient according to the description by Monson [3] (Fig. 2).

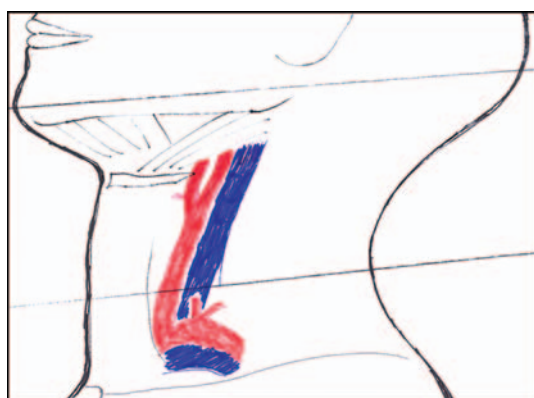


Fig. 2: Anatomic zones of the neck according to Monson (1). Zone I extends from the sternal notch and the level of the clavicles to the cricoid cartilage. From the cricoid cartilage to the angle of the mandible extends Zone II. Zone II extends from the angle of the mandible to the skull base

Tab. 1: Injured vessel in patients with penetrating neck trauma

Injured vessel	n
Common carotid artery	2
Internal carotid artery	2
External carotid artery	1
Thyroid artery	2
Vertebral artery	1
Internal jugular vein	7
External jugular vein	2

The vessels involved are shown in Table 1. The common carotid artery was injured in two patients, the internal carotid artery in two, the external carotid artery in one, the thyroid artery in two patients and the vertebral artery in 1 patient. In the arterial injuries vascular reconstruction was performed by venous interposition in one patient, venous patch plasty in one patient and direct suture in 6 patients (Fig. 3). Venous ligation was performed in all patients treated for venous injuries. Vessels involved were the internal jugular vein in 6 patients and the external jugular vein in three patients. In the 4 attempted suicides, all patients had lacerated the internal jugular vein, and one patient additionally the internal carotid artery. All patients with attempted suicides survived. Mortality in our patient group was 11.8% (2/17 patients). On the admission day a 20-year-old polytraumatized man died due to hemorrhagic shock. A 43-year-old man after staple gun trauma to the carotid bifurcation suffered an ipsilateral stroke and died on postoperative day 7. No additional neurological complications were observed.

Discussion

The optimal diagnostic and therapeutic approaches in the management of penetrating neck injuries have undergone considerable debate over the last years. As a vascular surgery unit we cannot give data on non-surgical treatment. Therefore, this report describes our operative results in comparison to the literature.

The trauma mechanism is important for the diagnostic and therapeutic strategy. In different patient age groups and geographical locations the trauma is caused by very different entities. In pediatric patients in Florida according to Abujamra the major cause of injury are motor vehicle accidents followed by gunshot wounds and animal bites [4]. Additionally there are major differences on trauma extent depending on a civilian or military background of the injured patient because of the energy transmitted by the weapon or missile. Kesser reported that stabbings have lower surrounding tissue injury than small calibre gunshot wounds which have lower tissue damage than high-powered rifle or shotgun blasts [5]. In contrast to the literature we saw no injuries to the neck from penetrating missiles, neither low-velocity nor high-velocity shots as described in 26% of patients by Bagheri

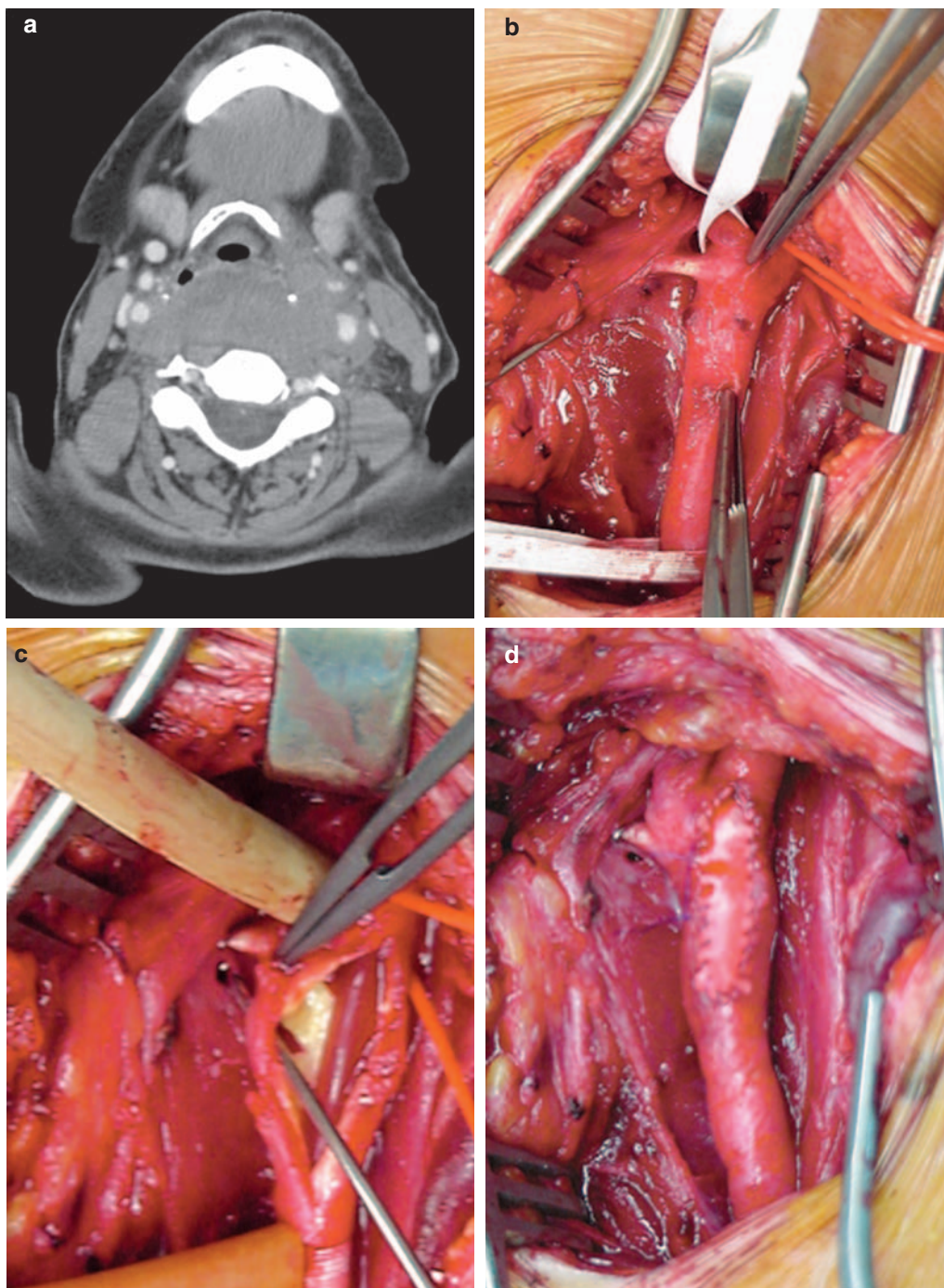


Fig. 3: Injury to distal common carotid artery in a 46-year-old female patient (knife, husband) (a): Preoperative CT-Angiography. (b) Intraoperative situs: no acute bleeding of the lesion without cross-clamping of the arteries. (c) After placement of the temporary intraluminal shunt and arteriotomy the transmural lesion is demonstrated. (d) Result after reconstruction by patch

[1]. Depending on the trauma mechanism injuries to the trachea the larynx, pharynx, esophagus and major nerves were described [5–7]. In our patient group the majority of stab wounds were inflicted by knives, there was no injury caused by a shot. This has contributed to the fact that we did not see the additional lesions in contrast to patient series as reported by Demetriades with 43% gunshot plus 4% shotgun injuries [7].

The majority of injuries in zone II in our patients group are comparable with the literature [9–11]. Specifically for zone II injuries the importance of a thorough physical examination to detect vascular injury was shown by Sekharan [12]. In the presence of hard signs including active bleeding, expanding hematoma, a bruit or thrill over the wound pulse deficit, and a central neurologic deficit corresponding to the side of the injury the patients

were taken directly to the operating room. 90% of these patients had either major arterial or venous injuries requiring repair [12]. According to Kesser's review zone II injuries are typically explored whereas the surgical complexity in approaching zones I and III has sparked a more selective approach to dealing with the injuries [5].

Both mandatory surgical exploration and selective approaches to the evaluation of patients with penetrating neck trauma have strong supporters in the literature. In diagnostic modalities the traditionally used invasive arterial angiography has been compared to high quality CT angiogram (CTA) with the advantage of the less invasivity in CTA and similar accuracy in diagnosis [13]. CTA provides information about the vascular structures as well as the soft tissue of the neck in a rapid way [14]. In recent studies the use of CTA could reduce the number of surgical neck explorations [11, 15]. Observational treatment as reported by Inaba for 8 hemodynamically stable patients with injury to the internal jugular vein out of 63 penetrating neck injuries has not been our treatment policy [16]. In all our patients active bleeding was indication for immediate surgery. 12 of 17 patients did not undergo diagnostic procedures. Bell who reported a large patient group and recommended selective surgical intervention stated as indications for immediate surgical intervention exsanguinating hemorrhage, expanding hematoma, shock, airway compromise and massive subcutaneous emphysema [10]. In our patient group we saw a man with a minor lesion to the thyroid artery leading to a massive hematoma with airway shift who needed bleeding control as well as removal of the hematoma (Fig. 4). In summary the decision between primarily surgical and a selective/observational approach remains highly controversial in the literature [17, 18].

In operative treatment there are also controversial issues. A recommendation is given by Maier for patients with injuries to the common or internal carotid artery to treat the arterial lesion by direct suture [19]. We lost a 43-year-old man because of carotid surgery without exploration of the intimal layer of the internal carotid artery due to postoperative carotid occlusion and fatal hemi-

spheric stroke. In patients with zone II injuries and stable cardiopulmonary function we now prefer complete exploration of the internal and distal common carotid artery including placement of a temporary intraluminal shunt to either safely exclude or to treat intimal lesions.

In the recent literature endovascular treatment of vascular injuries is discussed often. Interventional endoluminal treatment for vascular injuries was primarily applied to elective procedures treating posttraumatic pseudoaneurysms and occlusion of arteriovenous fistulas [20, 21]. Recently endovascular procedures were performed in emergency situations and even in hemodynamically unstable patients [22, 23]. After failed surgery with soft tissue trauma that did not allow complete vascular reconstruction or bleeding control endovascular treatment is a valid option [10, 24]. The still evolving endovascular technology may be of advantage for the patient in selected cases, but at present has little role for more significant injuries [1]. In our hospital the interventional emergency treatment is a difficult approach because an experienced interventional radiologist is not always in the hospital as opposed to the Board certified vascular surgery specialist on in-hospital call 24/7.

The mortality in our patient group with 11.8% is about as high as in comparable European centers with 12.5% in the higher range compared to large US-American trauma centers with 2.7% mortality in 223 patients in Los Angeles, 2.5% in Bronx/N.Y. [7, 8, 25]. A possible explanation for this is the fact that arterial injuries contribute substantially to negative outcomes in penetrating neck trauma, especially when the carotid arteries are involved [14, 26]. In our patients the percentage of arterial injuries with 47.6% was higher compared to 20% arterial lesions in Bell's patients [10]. Especially our patient who died of fatal ipsilateral stroke after staple-gun injury to the internal carotid artery might have been saved by an even more aggressive vascular surgical approach with exploration of the carotid after arteriotomy and inset of the intimal layer. Additionally the majority of our patients (9/17 = 53%) were transported to this tertiary care center by helicopter. A pre-hospital patient selection of more severely injured patients can have contributed additionally to the relatively high mortality. To improve patient care and treatment results in trauma victims currently a new trauma center is in the planning phase offering optimal diagnostic facilities and hybrid structures to prevent transportation times between ER, diagnostic facilities and operating suites.

Conclusion

Penetrating trauma to the neck is a serious and potentially lethal injury. CT-Angiography is recommended in the literature in diagnostic exploration of hemodynamically stable patients. Emergency surgery by an experienced vascular surgeon can help to determine the extent of trauma and its treatment. In arterial trauma to the carotid arteries a complete intraoperative exposure with external and internal inspection of the vessel including the intimal layer is recommended. The role of



Fig. 4: CT-Angiography immediately after admission in an intubated 66-year-old male patient showing major lateral displacement of the trachea by hematoma. The inferior thyroid artery was severed by after penetrating metal car fragments in a motor vehicle accident

endovascular treatment for vascular injury due to penetrating neck trauma needs further evaluation.

Conflict of interest

The authors declare that there is no conflict of interest.

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